BM25-RAG: A Sparse-Retrieval Baseline for Academic Question Answering

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Abstract

We describe BM25-RAG, a minimal retrieval-augmented baseline for academic question answering. The system splits documents into fixed 512-token chunks, retrieves the top six chunks with BM25, and asks a generator to answer with page citations. On the same synthetic benchmarks used by LightRAG, BM25-RAG reaches an F1 of 0.69 and a citation accuracy of 0.61 with an average latency of 0.9 seconds. It is fast, interpretable, and requires no GPU for retrieval, but it struggles when the question paraphrases the source or uses a synonym instead of the exact term. The baseline is useful for measuring how much a more elaborate hybrid pipeline actually contributes.

Keywords

BM25; sparse retrieval; retrieval-augmented generation; baseline; academic QA

1. Introduction

Retrieval-augmented generation is often evaluated with a dense retriever. A sparse baseline is equally important: it is cheap, transparent, and surprisingly strong on questions that contain exact terminology. This report documents such a baseline so that the contribution of hybrid retrieval and reranking can be measured against something simple rather than against nothing.

2. Method

Documents are split into fixed 512-token chunks with no page-aware boundaries. The retriever scores every chunk with BM25:

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BM25(q, d) = sum_t IDF(t) * (tf * (k1 + 1)) / (tf + k1 * (1 - b + b * |d| / avgdl))
```

We use k1 = 1.5 and b = 0.75, the standard defaults. The top six chunks are placed in the prompt with their page numbers, and the generator is asked to cite a page for every factual claim. No reranker and no dense retriever are used. The design deliberately favors simplicity over accuracy so that its failure modes are easy to interpret.

3. Experiments

We evaluate on SciQA-Demo and PaperQA-Demo, the same synthetic benchmarks used in the LightRAG report. Metrics are token-level F1, citation accuracy (CA), and average latency. All systems use the same six-passage context budget and the same generator.

Table 1: Sparse baseline versus hybrid system

Method	F1	CA	Latency
BM25-RAG	0.69	0.61	0.9 s
Dense-only baseline	0.74	0.66	1.3 s
LightRAG (hybrid + rerank)	0.82	0.79	1.8 s

BM25-RAG is 0.9 seconds faster than LightRAG but loses 0.13 F1 and 0.18 citation accuracy. The gap is concentrated in questions that use synonyms or paraphrase the source. On questions containing an exact model name or dataset name, the gap narrows to 0.03 F1.

3.1 Where the baseline wins

BM25-RAG is competitive on exact-term queries and on documents with highly distinctive vocabulary. It is also the easiest system to debug: a developer can inspect the matched terms and understand why a chunk was retrieved. For a small corpus with stable terminology, this baseline may be sufficient.

4. Discussion

The baseline has three strengths. It runs on CPU, its retrieval scores are interpretable, and it handles rare exact terms well. It also has three weaknesses. First, vocabulary mismatch is fatal: if the question says "retrieval quality" and the paper says "ranking effectiveness", BM25 may miss the passage. Second, fixed-size chunking breaks tables and formulas across boundaries. Third, the lack of a reranker leaves the generator with no second chance to reject a noisy candidate.

These tradeoffs suggest a simple deployment rule. If latency and hardware are the dominant constraints, start with BM25-RAG and add a reranker only if evaluation shows a clear gain. If answers must be citation-accurate across paraphrased questions, a hybrid pipeline is worth the additional 0.9 seconds.

4.1 Threats to Validity

The benchmarks are synthetic and English-only. The reported numbers illustrate the behavior of the pipelines rather than the performance of a production system. A real evaluation should include domain experts, naturally occurring questions, and documents with noisy OCR.

5. Conclusion

BM25-RAG is a useful lower bound for academic question answering. It is fast and transparent, and it establishes how much a hybrid retrieval and reranking pipeline must improve before its extra complexity is justified. On the synthetic benchmarks, that improvement is substantial for paraphrased questions and citation accuracy.

Limitations

- English-only synthetic benchmark; no human evaluation.
- Fixed-size chunking can split tables and formulas.
- No semantic matching; synonym and paraphrase queries are fragile.
- Citation accuracy depends on the quality of page metadata.

References (fictional)

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